Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll No.....

Name of the Program: B.Tech ECE Semester: V
Name of the Course: Electronics & Communication Engineering
Course Code: TEC-502
Paper Name: Digital System Design using VHDL

Time: 1:30 Hours **MM: 50**

Q.1	**(10 Marks)**	CO1
a)	Write the syntax for Entity and Architecture in VHDLand explain the VHDL code for full subtractor.	
	OR	
b)	Define Libraries in VHDL. Write VHDL code for NAND, NOR and XOR operation of two inputs.	
Q.2	**(10 Marks)**	CO1
a)	What is the significance of VHDL in digital circuit design. Explain the flow diagram of VHDL with the help of example.	
	OR	
b)	Explain the role of EDA tools in digital system design. Explain five pre-defined data types with examples.	
Q.3	**(10 Marks)**	CO2
a)	Explain Arithmetic operators Relational operators	
	OR	
b)	Explain signal and data attributes with the help of example. Explain Subtypes with the help of example.	
Q.4	**(10 Marks)**	CO1,CO2
a)	Explain the truth table of full subtractor, write the VHDL code for it and implement using basic gates.	
	OR	
b)	Explain1D, 2D 1D*1D Arrays with the help of example. Explain the syntax of GENERIC statement in VHDL.	
Q.5	**(10 Marks)**	CO2
a)	Explain data flow, behavioral and structural model for VHDL coding with the help of example.	

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	OR	
b)	SIGNAL a: BIT; SIGNAL b: BIT_VECTOR (7 DOWNTO 0); SIGNAL c: STD_LOGIC; SIGNAL d: STD_LOGIC_VECTOR (7 DOWNTO 0); SIGNAL e: INTEGER RANGE 0 TO 255; For the given conditions comment on the following statements (True/False) (i) a <= b (3); (ii) b (0) <= c; (iii) c <= d (5); (iv) a <= e; (v) b <= a;	